

01 · PROBLEM & DATA

Predicting the outcome of NBA possessions

RESEARCH QUESTION

To what extent can the outcome of an individual NBA possession be predicted from optical tracking data?

Possession-level 5-class classification

made_2 · made_3 · missed_shot · turnover · free_throws

One label per possession

Possessions are segmented from play-by-play; the terminal event defines the class. The full sequence of tracking frames up to that event is the input.

Team & scope

Nicolas Moy · Abhinav Surendra · Mostafa Dewidar. Three methods compared on one unified dataset.

DATASET

SportVU optical tracking · 2015-16 NBA season

636

regular-season games

72k

usable possessions

25 Hz

player + ball tracking

60/20/20

game-level train/val/test

Class distribution across usable possessions — note made_3 is rarest (~9%)

missed_shot

made_2

turnover

free_throws

made_3

02 · APPROACH

Three methods, one dataset, one held-out test split

METHOD A

BASELINE

XGBoost

Tabular baseline

- Per-possession summary features (min/max/mean/std over frames)
- Gradient-boosted decision trees
- Interpretable via gain importance

RATIONALE

Strongest baseline — handles mixed features, fast to train, tells us which aggregates matter.

METHOD B

COMPLETE

LSTM

Temporal sequence model

- Per-frame feature sequence
- Input encoder → 2× Conv1d
- BiLSTM + additive attention pool
- Trained with horizon augmentation

RATIONALE

Captures within-possession dynamics; enables early-prediction evaluation at any horizon.

METHOD C

BY FINAL

Graph Neural Net

Spatial relations

- Graph of 10 players + ball per frame
- Message passing over spatial edges
- Temporal aggregation over frames
- Captures formations & spacing

RATIONALE

Stretch goal from the proposal. Hypothesis: explicit spatial structure should help made_3 / zone-sensitive outcomes.

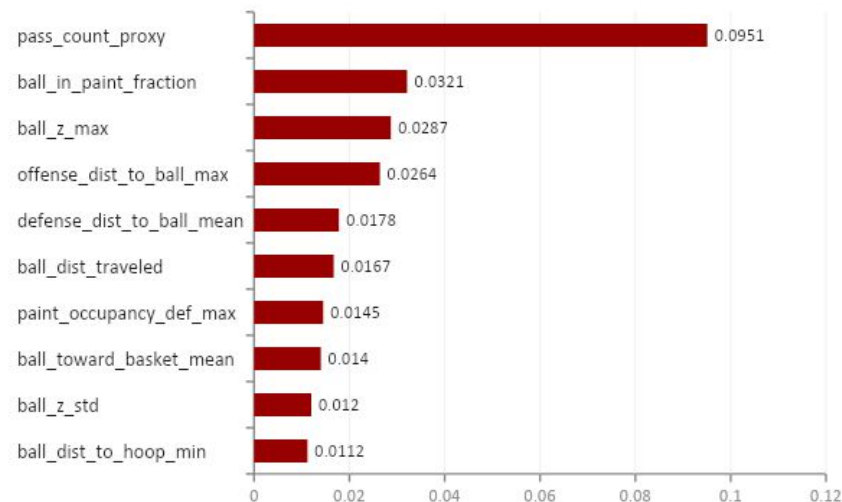
Shared across methods: same 636-game corpus · same 60/20/20 game-level split · same 5-class label scheme · same test set, held out until final eval.

03 · XGBOOST BASELINE

What engineered features drive the prediction?

TOP ENGINEERED FEATURES

Gain importance, top 10 of 200+

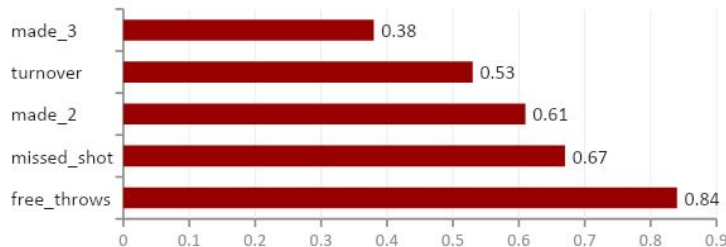


TEST ACCURACY

0.623

macro-F1 0.60 · n = 3,471

Per-class test F1



TAKEAWAY

A `pass\_count\_proxy` feature alone carries ~10% of the gain. Ball-movement aggregates (height, distance, paint-time) dominate. Floor-spacing features (centroid_y, possession_number) contribute almost nothing — candidates to drop.

04 · LSTM RESULTS

Conv → BiLSTM + Attention

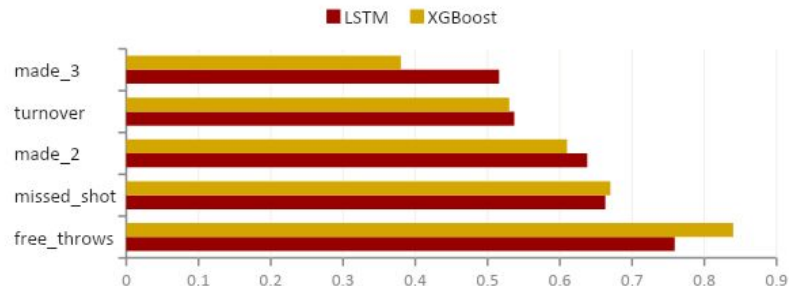
FULL-HORIZON MACRO-F1

0.623 ± 0.004*accuracy 0.631 ± 0.003 · seed-level CI is tight*

16-SECOND HORIZON

85.1% of full*macro-F1 0.530 ± 0.003 using only the first 16 of 24 seconds*

Per-class test F1 (LSTM vs. XGBoost)



LSTM confusion matrix (row-normalized)

	m_2	m_3	miss	TO	FT
m_2	0.59	0.07	0.14	0.14	0.05
m_3	0.15	0.56	0.13	0.14	0.02
miss	0.12	0.06	0.62	0.17	0.03
TO	0.10	0.04	0.16	0.66	0.03
FT	0.07	0.03	0.08	0.07	0.76

*rows = true · cols = predicted · made_3 → made_2 leak = 15%***LSTM vs. XGBoost: essentially tied at full horizon (+1 pt accuracy, +2 pts macro-F1). The temporal model's real win is early-prediction — 85% of signal recoverable by 16s.**

What's been hard so far — and what's next

Compute & disk pressure

SOLVED

Raw SportVU JSON is ~50 GB; packed possession arrays ~3 GB. Solved with Colab Pro (A100) + Drive mount + memory-mapped .npy arrays — constant-RAM training across all 72k possessions.

Possession segmentation

SOLVED

Raw tracking has no possession boundaries. Built rules-based segmenter from play-by-play: offense-change + hard-terminal events, with filters on invalid-frame ratio and missing shot clock. Yields ~72k usable possessions from ~130k raw.

GNN design & fair comparison

NEXT

Next: implement a spatial GNN (player+ball nodes, distance-weighted edges) with temporal aggregation. Match input features and training budget to the LSTM so the comparison is clean. Target: beat LSTM on made_3 specifically.